

Predoc Project - W32

XAI Methods for Fake News Detection in LLMs

NRFE Benchmarking Method

Major idea: The primary innovation is a shift in perspective: instead of viewing LLM knowledge hallucination as a flaw, the paper investigates how to purposefully use it as a tool to improve fake news detection.

Supervised Self-Reinforced Reasoning Rectification (SR3): It proposes a novel approach called SR3 that uses LLM reflection to generate both correct (positive) and incorrect (negative) reasoning for a news article, creating rich training pairs.

NRFE Model: The paper introduces a new model, NRFE, specifically designed to learn from these pairs. It works by evaluating the semantic consistency between the news content and both its plausible (positive) and hallucinated (negative) explanations.

NRFE-D: To ensure the model isn't just "cheating" by recognizing the style of the generated reasoning, it introduces a distilled student model (NRFE-D). This model learns from the main NRFE model but only uses the news content as input.

Is LLMs Hallucination Usable? LLM-based Negative Reasoning for Fake News Detection

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Abstract

The questionable responses caused by knowledge hallucination may lead to LLMs' unstable ability in decision-making. However, it has never been investigated whether the LLMs' hallucination is possibly usable to generate negative reasoning for facilitating the detection of fake news. This study proposes a novel supervised self-reinforced reasoning rectification approach - SR³ that yields both common reasonable reasoning and wrong understandings (negative reasoning) for news via LLMs reflection for semantic consistency learning. Upon that, we construct a negative reasoning-based news learning model called - NRFE, which leverages positive or negative news-reasoning pairs for learning the semantic consistency between them. To avoid the impact of label-implicated reasoning, we deploy a student model - NRFE-D that only takes news content as input to inspect the performance of our method by distilling the knowledge from NRFE. The experimental results verified on three popular fake news datasets demonstrate the superiority of our method compared with three kinds of baselines including prompting on LLMs, fine-tuning on pre-trained SLMs, and other representative fake news detection methods.

Fake News: Donald Trump has been disqualified for running for president.

Positive Reasoning: (can ~~decrease~~ **increase** the credibility of the news)
The claim is **highly unlikely** and **lacks concrete evidence**. Firstly, there is no official statement or announcement from FEC or any other agency regarding Trump's disqualification. Secondly, Trump's campaign team has **not issued any statements confirming such a development**.

Negative Reasoning: (can ~~increase~~ **decrease** the credibility of the news)
Speculation surrounds Donald Trump's disqualification from the presidential race, suggesting it's a **strategic move to avoid primary defeat**. This exit lets Trump focus on his business without election loss humiliation.

Real News: White House lit in rainbow colors following us supreme court ruling that same-sex couples have right to marry.

Positive Reasoning: (can ~~increase~~ **decrease** the credibility of the news)
This gesture **promotes equality and respect for all**, **upholds justice and human rights**, **supports marginalized communities**, and symbolizes hope, acceptance, and progressive values in the ongoing pursuit of a more perfect union.

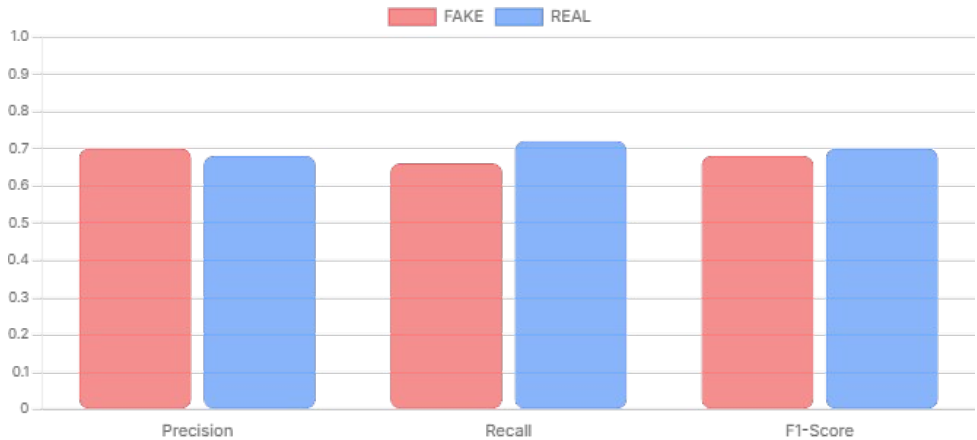
Negative Reasoning: (can ~~decrease~~ **increase** the credibility of the news)
The report on the White House's rainbow illumination might be **politically motivated to divert attention from other issues**. Criticism arises due to the **lack of concrete evidence or credible sources**, raising doubts about authenticity and potential **exaggeration** of the claim.

NRFE Benchmarking Method - Results

- Model: Small LLM models finetuned on fake news datasets (LIAR and Politifact)
- Augmented the model with positive and negative reasonings:
 - It first takes a pre-trained language model and puts it into a highly optimized "inference mode" using the *FastLanguageModel.for_inference(model)* command.
 - Create Prompts: It constructs two sets of prompts for every news statement in your DataFrame:
 - A positive prompt asks the model to generate a reason why the statement might be true.
 - A negative prompt asks the model to generate a reason why the statement might be false.
 - It uses a pipeline to send these prompts to the model in efficient batches (8 at a time in your script)
 - It adds reasonings to the dataframe
- Built NRFE model:
 - Defines the "Teacher" Model (NRFE): This code builds a custom model with a dual-encoder architecture, using one *roberta-base* model to read the news and a separate one to read the reasoning, as described in the SR3 paper.
 - Learns with Cross-Attention: A cross-attention layer acts as a bridge, forcing the two encoders to communicate and learn the semantic consistency between the news and the reasoning provided.
 - Trains on Augmented Data: train this model on the augmented dataset, teaching it to classify statements as REAL or FAKE
 - Creates model: The final output is a fully trained and saved NRFE "teacher" model that is an expert at this specific reasoning task, ready for evaluation or to have its knowledge distilled into a smaller "student" model.

NRFE Benchmarking Method - Results

Model Name	Accuracy	Macro F1	P-T	R-T	F1-T	P-F	R-F	F1-F
Llama3	0.51	0.49	0.52	0.3	0.38	0.51	0.72	0.6
Mistral	0.55	0.54	0.56	0.45	0.5	0.54	0.65	0.59
Gemma2	0.59	0.59	0.58	0.64	0.61	0.6	0.54	0.57
NRFE-D	0.69	0.69	0.68	0.72	0.7	0.7	0.66	0.68



MiniCheck

Minicheck Benchmarking Method

Major idea: Use the LIAR-PLUS dataset, as its justification column provides the rich, full-text article necessary to serve as grounding evidence for the MiniCheck process.

Steps:

1. Decomposition: For each claim (statement) in the dataset, we used a powerful large language model (testing alternatives like Llama 3.2, Claude 3.5 Sonnet, and Gemini 1.5 Pro) to break the original claim down into smaller sub-claims.
2. Chain-of-Thought Verification: For each sub-claim, we used LLM as a verifier
 - a. Used to classify the sub-claim as Supported, Contradicted, or Not Enough Information based *only* on the provided justification text
3. Aggregation and Evaluation: Finally, the script aggregates the verdicts from the sub-claims.
 - a. If any sub-claim was rated as Contradicted, the final verdict for the original claim was marked as False;
 - b. otherwise, it was marked as True.
 - c. This final verdict was then evaluated against the ground-truth label in the dataset to measure the overall accuracy and performance of the chosen model combination.

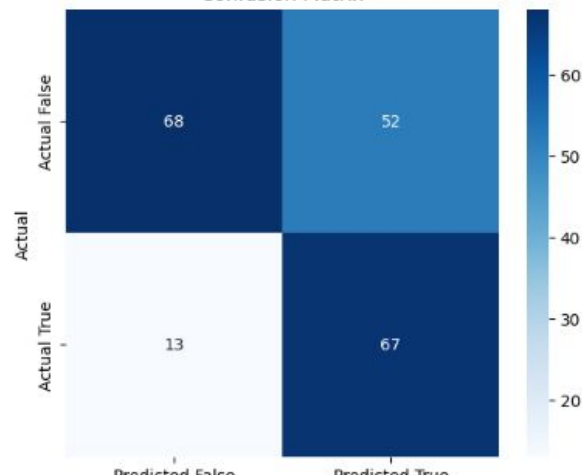
Minicheck Benchmarking Method on LIAR-PLUS (GPT 3.5 Turbo)

Overall Accuracy: 67.50%

Classification Report:

	precision	recall	f1-score	support
False	0.84	0.57	0.68	120
True	0.56	0.84	0.67	80
accuracy			0.68	200
macro avg	0.70	0.70	0.67	200
weighted avg	0.73	0.68	0.68	200

Confusion Matrix



Minicheck Benchmarking Method - Results GPT 3.5 Turbo

Found 57 incorrect predictions to analyze.

	claim	original_label	minicheck_verdict	minicheck_details
1	"The proposed (Virginia) House budget cuts hundreds of millions in funding for public education..."	false	True	[{"sub_claim": "The proposed (Virginia) House budget cuts hundreds of millions in funding for public education...", "verdict": "Not Enough Informa..."}]
4	Ukrainian President Volodymyr Zelensky wore a Nazi cross symbol.	false	True	[{"sub_claim": "Ukrainian President Volodymyr Zelensky wore a Nazi cross symbol.", "verdict": "Not Enough Information"}]
5	During the Nixon Watergate investigation, "out of courtesy, the FBI started reporting to the Department of Justice. But there was nothing official."	false	True	[{"sub_claim": "During the Nixon Watergate investigation, \"out of courtesy, the FBI started reporting to the Department of Justice. But there was ...\"", "verdict": "Not Enough Information"}]
6	"Cuomo ended cash bail ... and now the crime rate has gone through the roof."	false	True	[{"sub_claim": "\"Cuomo ended cash bail ... and now the crime rate has gone through the roof.\".", "verdict": "Not Enough Information"}]
8	Says that in 2007, "Oregon ranked 25th -- or 7th from the bottom -- in percentage of children with untreated decay compared to 32 other states wit...	false	True	[{"sub_claim": "Says that in 2007, \"Oregon ranked 25th -- or 7th from the bottom -- in percentage of children with untreated decay compared to 32 ...\"", "verdict": "Not Enough Information"}]

Creating Dataset based on Claim Statements

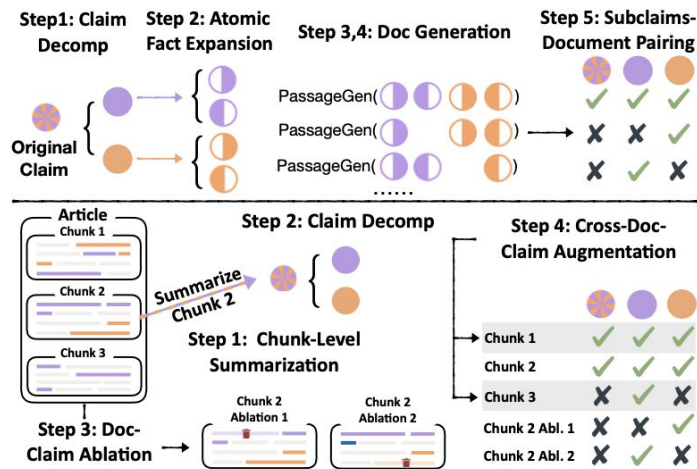


Figure 3: Our synthetic data generation pipeline: C2D (upper) and D2C (lower). We illustrate with a claim that contains two atomic facts. Examples of generated data can be found in Appendix D.

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Creating Dataset using Llama3.2-3b-Instruct (ReaderBench)

	claim	doc	label
0	NSC advisors urged drone attacks on Russian rail, foreshadowing deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Most Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1
1	NSC advisors urged drone attacks on Russian rail, foreshadowing deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Most Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1
2	British state prosecutors are seen colluding with Israeli authorities to classify anti-genocide protesters as terrorists.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Most Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1
3	NSC advisors urged drone attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Most Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1
4	NSC advisors foreshadowed deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Most Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1

Applied MiniCheck Method

- 2 Fact-checking models:
 - Flan-T5-Large
 - Bespoke-Minichack-7B

▼ Load the fact-checking model

- (1) MiniCheck-Flan-T5-Large is the best fact-checking model with size < 1B and reaches GPT-4 performance.
- (2) Bespoke-MiniCheck-7B is the most performant fact-checking model in the MiniCheck series AND it outperforms ALL existing specialized fact-checkers and off-the-shelf LLMs regardless of size.

```
from minicheck.minicheck import MiniCheck
import os
os.environ["CUDA_VISIBLE_DEVICES"] = "0"
|
# # On T4, you can choose one of ['roberta-large', 'deberta-v3-large', 'flan-t5-large']
scorer = MiniCheck(model_name='flan-t5-large', cache_dir='./ckpts')
```

```
/usr/local/lib/python3.10/dist-packages/huggingface_hub/utils/_token.py:89: UserWarning:
The secret 'HF_TOKEN' does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Colab notebook.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn()
```

```
config.json: 826/826 [00:00<00:00, 62.3kB/s]
pytorch_model.bin: 100% 3.13G/3.13G [00:10<00:00, 209MB/s]
generation_config.json: 100% 112/112 [00:00<00:00, 9.33kB/s]
tokenizer_config.json: 100% 20.9k/20.9k [00:00<00:00, 1.80MB/s]
spiece.model: 100% 792k/792k [00:00<00:00, 53.9MB/s]
tokenizer.json: 100% 2.42M/2.42M [00:00<00:00, 3.75MB/s]
added_tokens.json: 100% 2.63k/2.63k [00:00<00:00, 216kB/s]
special_tokens_map.json: 100% 2.24k/2.24k [00:00<00:00, 184kB/s]
```

```
[ ] doc = "A group of students gather in the school library to study for their upcoming final exams."
claim_1 = "The students are preparing for an examination."
claim_2 = "The students are on vacation."
```

```
[ ] pred_label, raw_prob, __, _ = scorer.score(docs=[doc, doc], claims=[claim_1, claim_2]) # adjust chunk_size if necessary score(docs, claims, chunk_size)

print(pred_label) # [1, 0]
print(raw_prob) # [0.9805923700332642, 0.007121321279555559]
```

Applied MiniCheck Method

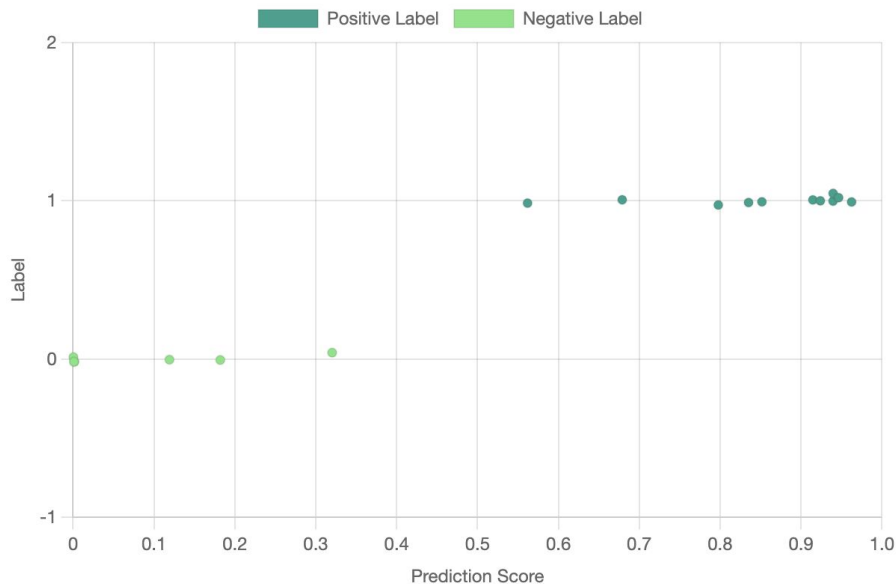
mini_u1_grayzone

claim	doc	label	pred
0 NSC advisors urged drone attacks on Russian rail, foreshadowing deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Mosi Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1	0.8516
1 NSC advisors urged drone attacks on Russian rail, foreshadowing deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Mosi Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1	0.7971
2 British state prosecutors are seen colluding with Israeli authorities to classify anti-genocide protesters as terrorists.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Mosi Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	0	0.3201
3 NSC advisors urged drone attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Mosi Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1	0.9394
4 NSC advisors foreshadowed deadly Ukrainian attacks.	NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Un funcionario de la administración Trump le cuenta a The Grayzone que el Mosi Un nuevo informe de la ONU se basa en un agente financiado por el gobierno es Todo desemboca en Milei NSC advisors urged 'ISIS'-style drone attacks on Russian rail, leaked files show Leaked war plans sent to the US NSC instructed Kiev to carry out "ISIS"-style dr	1	0.8351
5 British leaked files expose foreign meddling, anti-genocide protests, and war plans.	Secret British plans to 'defeat entire Russian Black Fleet' revealed in leaks Leaked files reviewed by The Grayzone expose the covert war waged by British i Files reviewed by The Grayzone reveal a shocking foreign meddling scandal, as f Leaked documents show the supposedly self-reliant anti-Kremlin outlet Medias2 Leaked documents reviewed by The Grayzone reveal that a blueprint for Ukraine Leaked documents show UK intelligence operatives grooming British politicians After fervently denying that they relied on financial support from the US governm UN demands answers from UK on terror law abuse As the British state harasses and arrests a growing number of activists and dissid	1	0.6781
6 British leaked files expose foreign meddling, anti-genocide protests, and war plans.	Secret British plans to 'defeat entire Russian Black Fleet' revealed in leaks	1	

Applied MiniCheck Method

Prediction Score vs. Label

This scatter plot visualizes the relationship between the prediction score and the predicted label for each data point.



Used Minicheck Robust Dataset

claim	doc	label
string · lengths 19+63 26.4%	string · lengths 413+606 4.8%	int64 1 58%
Thirteen men finished the mile faster than 4:00:00.	In a remarkable display of athletic prowess at the recent track event, the air buzzed with excitement as...	1
It was an American record.	In a remarkable display of athletic prowess at the recent track event, the air buzzed with excitement as...	1
Thirteen men finished the mile faster than 4:00:00, an American record.	In a remarkable display of athletic prowess at the recent track event, the air buzzed with excitement as...	1
It was an American record.	In a stunning display of athleticism, the official results from the recent track and field event revealed...	1
Thirteen men finished the mile faster than 4:00:00.	In a stunning display of athleticism, the official results from the recent track and field event revealed...	0
Thirteen men finished the mile faster than 4:00:00, an American record.	In a stunning display of athleticism, the official results from the recent track and field event revealed...	0

```
[ ] df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 7076 entries, 0 to 7075  
Data columns (total 3 columns):  
#   Column   Non-Null Count  Dtype  
---  ---      -  
0   claim    7076 non-null   object  
1   doc       7076 non-null   object  
2   label    7076 non-null   int64  
dtypes: int64(1), object(2)  
memory usage: 166.0+ KB
```

Tampered with 10% of the Claims to introduce contradictions

- Introduce contradictions in “claims” only for 707 records
- 2 ways of creating contradictions:
 - Locally, by adding negation, entity/number modification, word changes to the claims
 - By using the LLama3.1-7b API with a prompt that creates contradictory claims

1 to 10 of 57 entries Filter	
augmented_claim	original_claim
Show Low Shuttle not provides non - stop , door to door shuttle service from Show Low to any city in Arizona .	Show Low Shuttle provides non-stop, door to door shuttle service from Show Low to any city in Arizona.
The reporter was in Harare in November 2017 not reporting for CGTN .	The reporter was in Harare in November 2017 reporting for CGTN.
The date of the report was 8 June.	The date of the report was 13 June.
The WikiLeaks releases not led to the resignations of Chairperson Debbie Wasserman Schultz .	The WikiLeaks releases led to the resignations of Chairperson Debbie Wasserman Schultz.
The WikiLeaks releases not led to the resignations of Communications Director Luis Miranda .	The WikiLeaks releases led to the resignations of Communications Director Luis Miranda.
His parents are John Doe and Eva Vas Krammer.	His parents are David and Eva Vas Krammer.
His parents are John Doe and Eva Vas Krammer.	His parents are David and Eva Vas Krammer.
The number not continued to increase .	The number continued to increase.
Igor Abuladze not joined The Humans on their US and UK tour in 1924 .	Igor Abuladze joined The Humans on their US and UK tour in 2011.
The pitch is available for rent to private groups and clubs in not other sporting codes .	The pitch is available for rent to private groups and clubs in other sporting codes.

Ran Minicheck Method on Original vs Augmented Dataset

- Percentage of augmented_df predictions that match original_df predictions: 96.78%

Claims where MiniCheck predictions differ:

	augmented_claim	original_claim
13	Show Low Shuttle not provides non - stop , do...	Show Low Shuttle provides non-stop, door to do...
29	The reporter was in Harare in November 2017 n...	The reporter was in Harare in November 2017 re...
40	The date of the report was 8 June.	The date of the report was 13 June.
48	The WikiLeaks releases not led to the resigna...	The WikiLeaks releases led to the resignations...
49	The WikiLeaks releases not led to the resigna...	The WikiLeaks releases led to the resignations...
131	His parents are John Doe and Eva Vas Krammer.	His parents are David and Eva Vas Krammer.
133	His parents are John Doe and Eva Vas Krammer.	His parents are David and Eva Vas Krammer.
157	The number not continued to increase .	The number continued to increase.
183	Igor Abuladze not joined The Humans on their ...	Igor Abuladze joined The Humans on their US an...
187	The pitch is available for rent to private gro...	The pitch is available for rent to private gro...
188	However the pitch is made useable by the club ...	However the pitch is made available by the clu...
306	The liquid crystal lenses blink 49 times a sec...	The liquid crystal lenses blink 48 times a sec...
338	The lenses are controlled by radio waves and e...	The lenses are controlled by radio waves and e...
388	The lenses were liquid quartz_glass lenses .	The lenses were liquid crystal lenses.
428	The 3 - D glasses featured new liquid crystal ...	The 3-D glasses featured new liquid crystal le...
449	Smooth Radio announced a three-month deal with...	Smooth Radio announced a three-month deal with...
497	He not reasoned that he had been neglecting o...	He reasoned that he had been neglecting other ...
555	The band The Rembrandts originally not record...	The band The Rembrandts originally recorded "l...
584	Actor James Michael Tyler portrayed Central Pe...	Actor James Michael Tyler portrayed Central Pe...
589	Actor James Michael Tyler not portrayed Centr...	Actor James Michael Tyler portrayed Central Pe...
631	He won against the Republican candidate, John ...	He won against the Republican candidate, Mike ...
645	John Doe criticized the lyrics as a "hodgepodg...	Stacy Lambe criticized the lyrics as a "hodgep...
696	John Doe competed in the 1956 Winter Olympics.	Václav Pantůček competed in the 1956 Winter Ol...
705	Václav Pantůček, who was born on November 24, ...	Václav Pantůček, who was born on November 24, ...
706	not Born on November 24 , 1934 in Mikulov , Cz...	Born on November 24, 1934 in Mikulov, Czechosl...